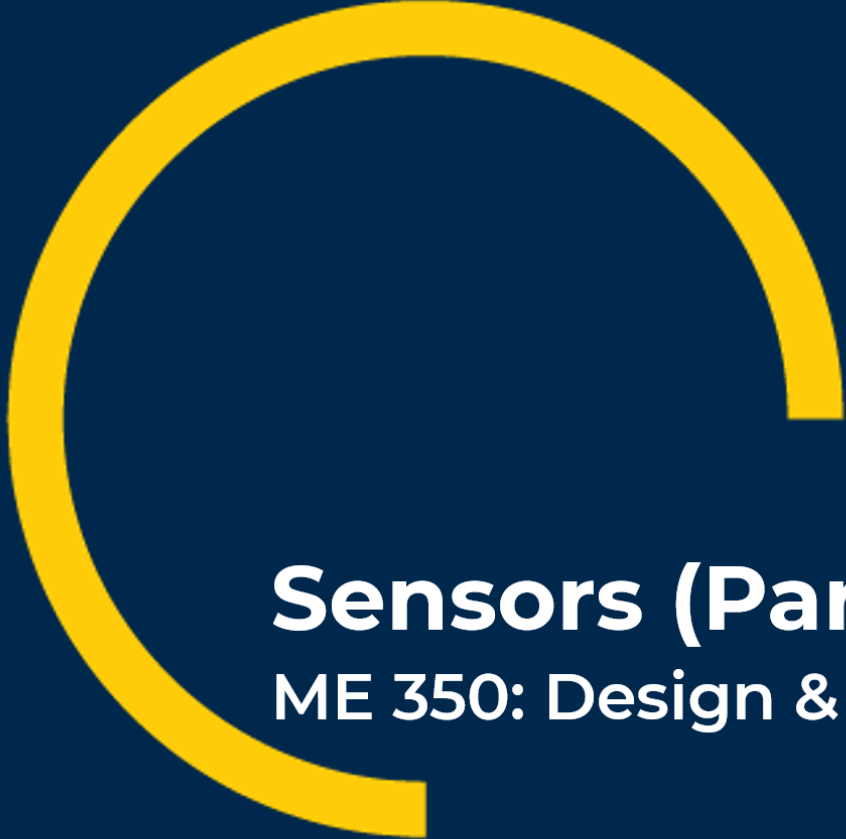




Sensors (Part 2)

ME 350: Design & Manufacturing II

Jeffrey Koller

Some slides adopted from Mike Umbriac, David Remy, Chinedum Okwudire, & Shorya Awtar



MECHANICAL
ENGINEERING

UNIVERSITY OF MICHIGAN

Announcements



- HW5 is Due tomorrow at 1:30 PM
- HW6 is set to be released today and Due Nov. 6th
 - There is content on signals in that homework that we will be discussing by the end of next week (10/30)

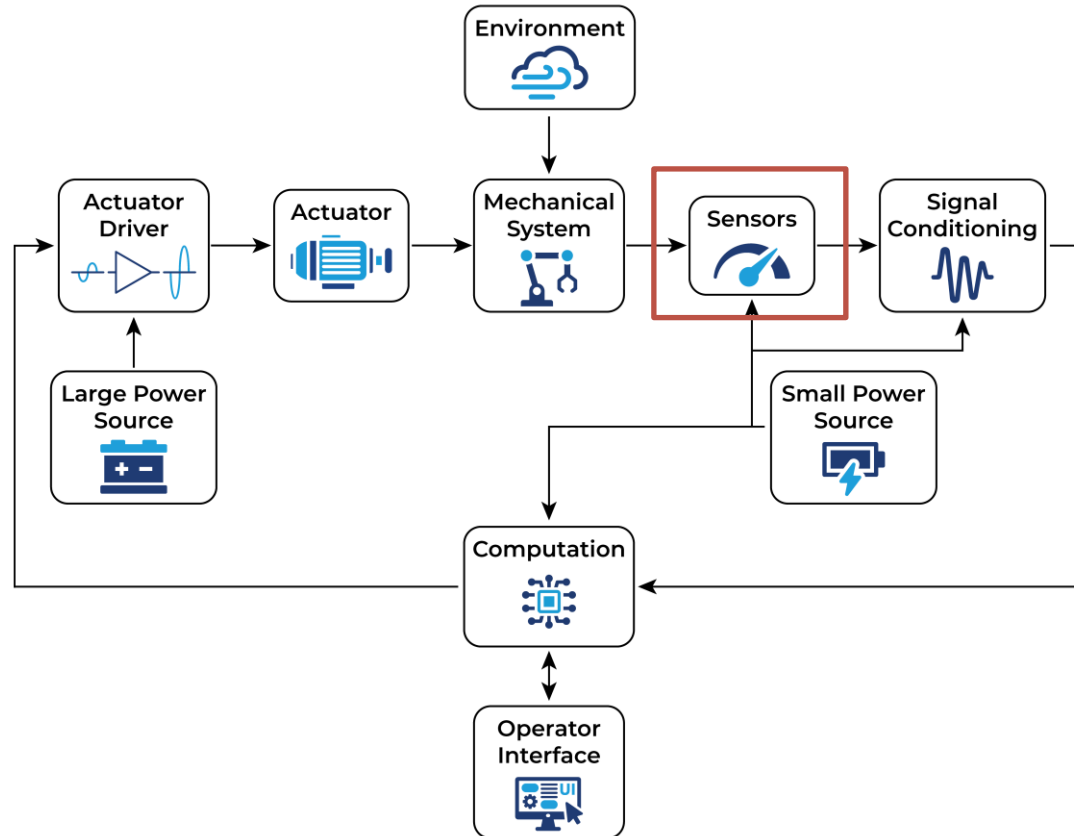
Today's Learning Objectives



By the end of today's class you should...

- Be able to understand how various analog sensors work
 - Potentiometer
 - Accelerometer
 - Rate Gyroscope
 - IR Proximity Sensors
 - Color Sensor
- Describe some of the key characteristics of sensors

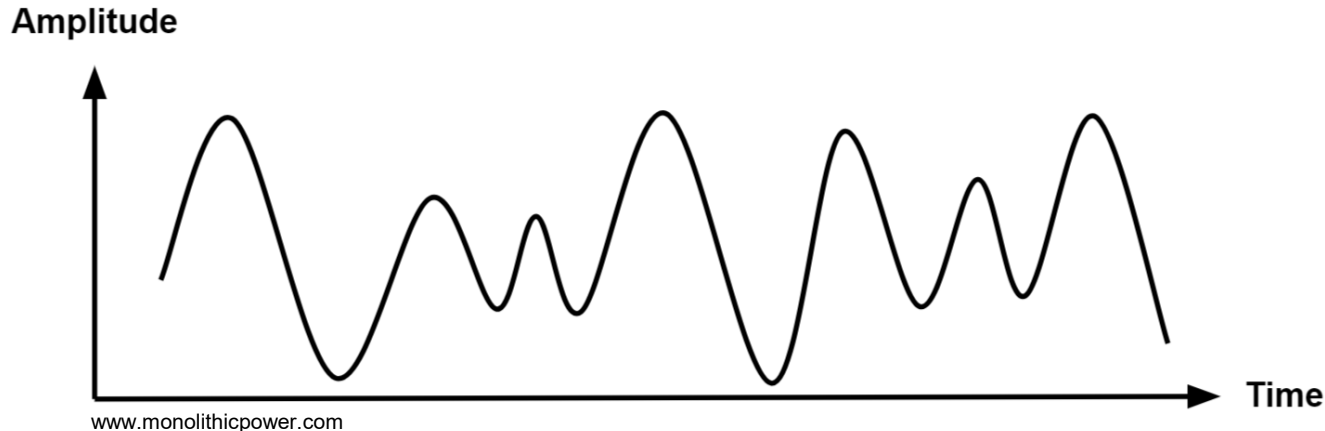
Context for Today's Lesson



Analog Sensors (Recap from Last Time)



- **Analog sensors** produce an output signal that varies **continuously** over time
- Ideally, the size of the analog output signal is **proportional** to the size of the variable being measured (i.e. they have a linear relationship)
- Unfortunately, **nonlinear** relationships are very common
- The output signal is typically either **current** or **voltage**



Analog Potentiometer

- A potentiometer is a **variable voltage divider**
- Produces a voltage that is **proportional** to angular (or linear) position of the wiper



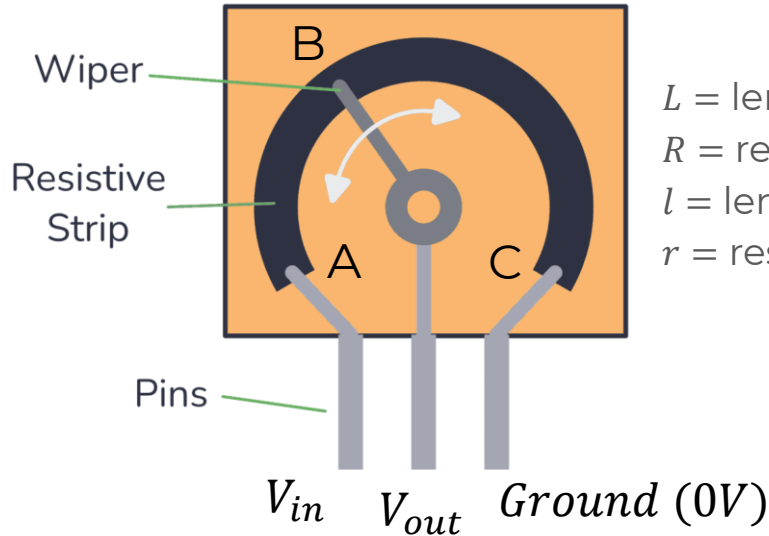
Angular



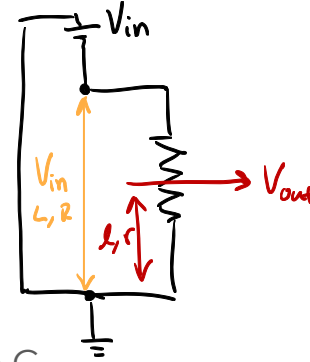
Linear

Analog Potentiometer

- A potentiometer is a **variable voltage divider**
- Produces a voltage that is **proportional** to angular (or linear) position of the wiper



L = length from A to C
 R = resistance from A to C
 l = length from B to C
 r = resistance from B to C



$$V_{in} = iR$$

$$V_{out} = ir$$

$$\frac{V_{out}}{V_{in}} = \frac{\cancel{i}r}{\cancel{i}R} = \frac{l}{L}$$

$$l = \frac{L}{V_{in}} V_{out}$$

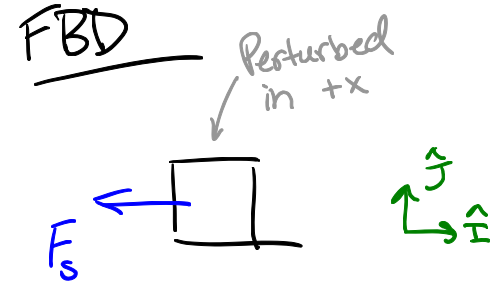
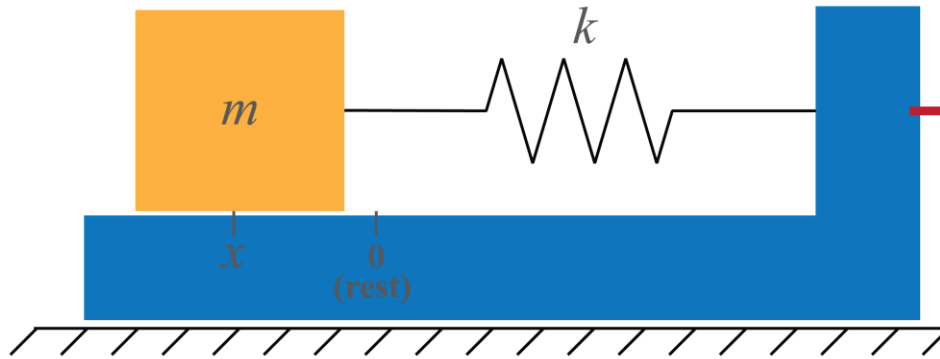
$$\therefore \frac{L}{V_{in}} \triangleq \frac{1}{S}$$

sensitivity

$$V_{out} = Sl$$

Inertial Sensors - Accelerometer

- Inertial sensors are sensors that work on the basis on **Newton's 2nd Law**
- Accelerometers are a great example of this:



| Newton's | $\Sigma \vec{F} = m\vec{a}$

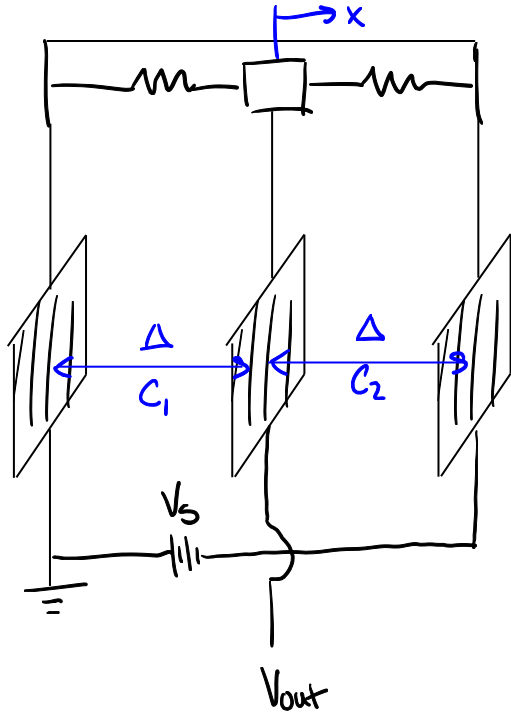
$\hat{i} / -kx = ma$

$$a = \frac{-kx}{m}$$

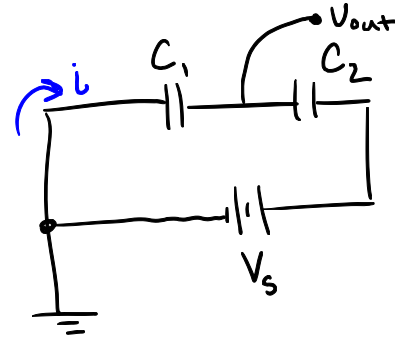
- If we can measure x then we know a !
- Measuring x : Capacitance, inductance, piezoelectric effect

Inertial Sensors - Accelerometer

Capacitance



$$\frac{V_{out}}{V_s} = ?$$



Using an impedance approach:

$$V_{out} = I(s) \left(\frac{1}{C_1 s} \right)$$

$$V_s = I(s) \left(\frac{1}{C_1 s} + \frac{1}{C_2 s} \right)$$

$$\begin{aligned} \frac{V_{out}}{V_s} &= \frac{\frac{1}{C_1 s}}{\frac{1}{C_1 s} + \frac{1}{C_2 s}} \\ &= \frac{C_2}{C_1 + C_2} \end{aligned}$$

Recall

$$C = \frac{\epsilon A}{d}$$

$$\text{So } C_1 = \frac{\epsilon A}{\Delta + x} \quad \& \quad C_2 = \frac{\epsilon A}{\Delta - x}$$

Inertial Sensors - Accelerometer

Plugging in:

$$\frac{V_{out}}{V_s} = \frac{\frac{\cancel{EA}}{\Delta - x}}{\frac{\cancel{EA}}{\Delta + x} + \frac{\cancel{EA}}{\Delta - x}} = \frac{\Delta + x}{2\Delta}$$

$$V_{out} = \left(\frac{1}{2} + x \frac{1}{2\Delta} \right) V_s$$

$$V_{out} = \underbrace{\frac{V_s}{2}}_{bias} - \underbrace{\frac{m V_s}{2k\Delta}}_{Sensitivity} a$$

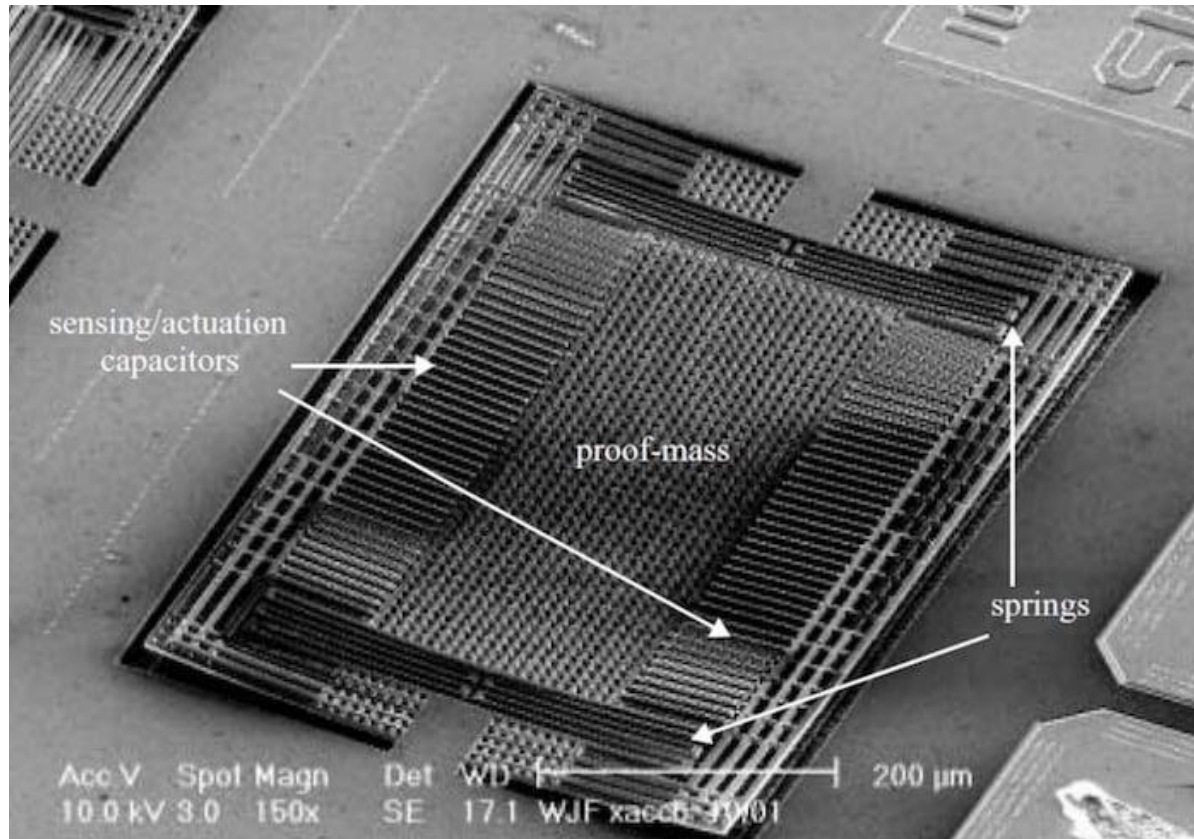
From above

$$a = -\frac{kx}{m} \text{ so } \dots$$

$$x = -\frac{am}{k}$$

$$\therefore V_{out} = V_b + S'a$$

Inertial Sensors - Accelerometer

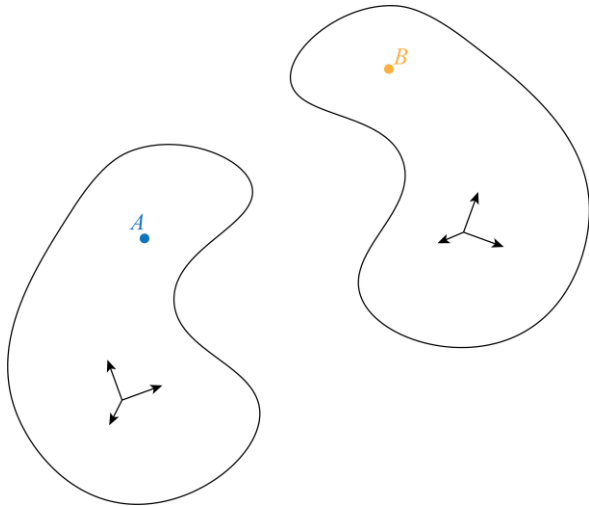


Inertial Sensors – Rate Gyro

- Another inertial sensor is the **rate gyroscope** which measures **angular velocity**
- This measurement is based upon the **Coriolis effect**
- Recall the kinematic equation for the acceleration of a point on a body relative to another moving body

$$\vec{a}_B = \vec{a}_A + \dot{\vec{\omega}}_A \times \vec{r}_{B/A} + \vec{\omega}_A \times (\vec{\omega}_A \times \vec{r}_{B/A}) + \underbrace{2\vec{v}_{rel} \times \vec{\omega}_A}_{\text{Coriolis Acceleration}} + \vec{a}_{rel}$$

Coriolis Acceleration



Inertial Sensors – Rate Gyro

$$\vec{a}_{cor,B} = 2\vec{v}_{rel} \times \vec{\omega}_A$$



Inertial Sensors – Rate Gyro

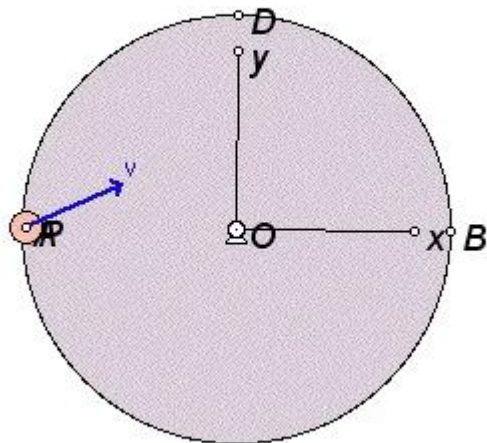
$$\vec{a}_{cor,B} = 2\vec{v}_{rel} \times \vec{\omega}_A$$



Inertial Sensors – Rate Gyro

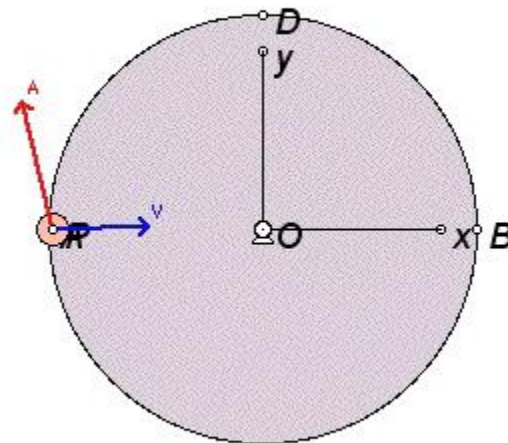
Merry-Go-Round Example

View by Observer FIXED to Ground



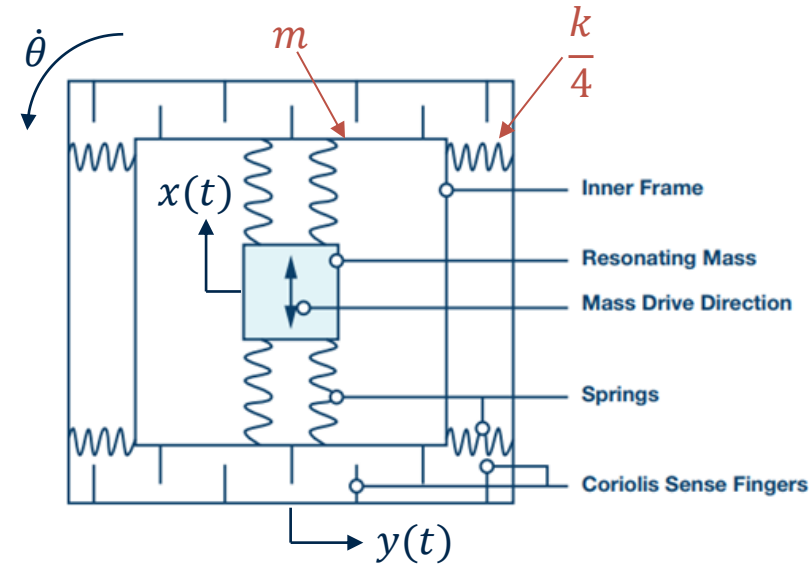
Merry-Go-Round Example

View by Observer MOVING with Merry-Go-Round

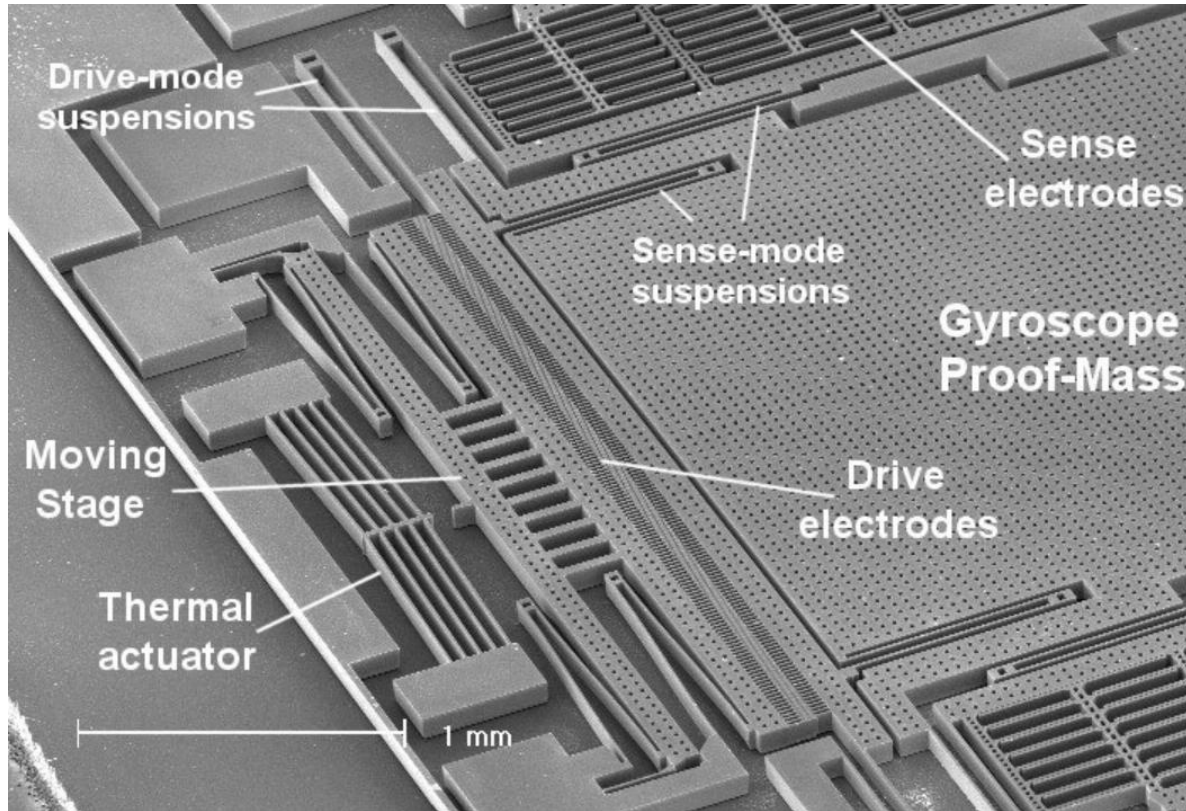


Inertial Sensors – Rate Gyro

- Drive the mass at a known oscillating velocity
- Any detected side to side motion is assumed to be due to the Coriolis acceleration meaning the body it is on is rotating!



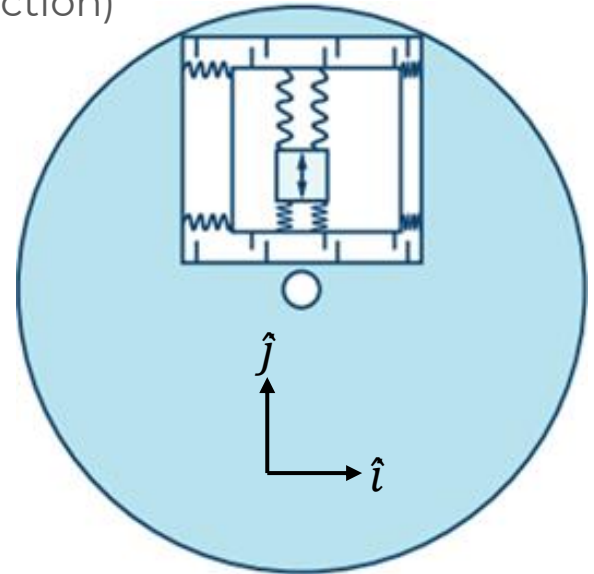
Inertial Sensors – Rate Gyro



Inertial Sensors – Rate Gyro

Detecting the direction of rotation:

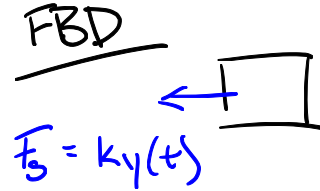
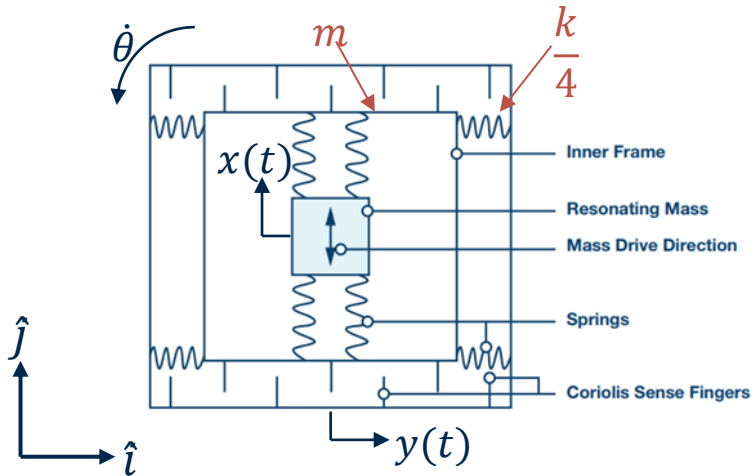
- Recall: $\vec{a}_{cor,B} = 2\vec{v}_{rel} \times \vec{\omega}_A$
- Let's fix the rate gyro to a body that can rotate
- At this instant, the proof mass has an downward velocity ($-\hat{j}$ direction)
- We observe that the inner frame moves right ($+\hat{i}$ direction)
- **What direction is body rotating?**



Inertial Sensors – Rate Gyro

Determining the angular rate ($|\vec{\omega}_A|$ or $\dot{\theta}$):

- Recall: $\vec{a}_{cor,B} = 2\vec{v}_{rel} \times \vec{\omega}_A$
- $x(t) = X \sin(\omega t)$: Displace of resonating mass within the inner frame
- $v(t)$: velocity of the resonating mass within the inner frame
- k : Total stiffness attached to inner frame
- m : Total mass of inner frame and contents
- $y(t)$: Displacement of inner frame within outer



Newton's law (scalar)

$$m a_{cor} = F_s = k y(t)$$

$$x(t) = X \sin(\omega t) \xrightarrow{d/dt} v(t) = \omega X \cos(\omega t)$$

$$|a_{cor}| = 2 v(t) \dot{\theta} = 2 \dot{\theta} \omega X \cos(\omega t)$$

$$m (2 \dot{\theta} \omega X \cos(\omega t)) = k y(t) \quad \text{measure}$$

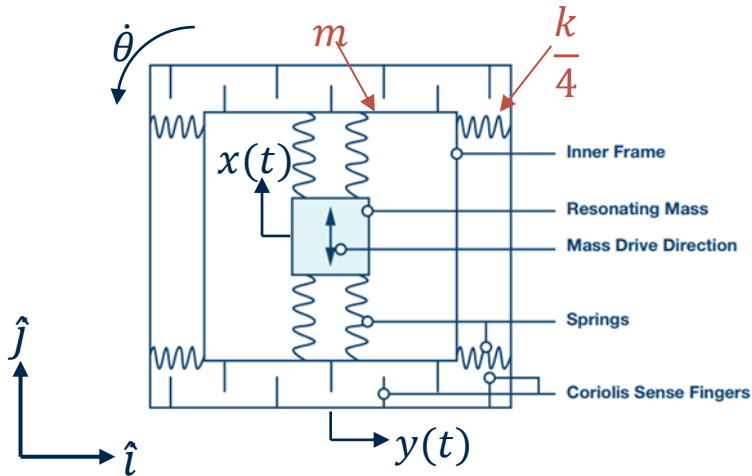
Inertial Sensors – Rate Gyro

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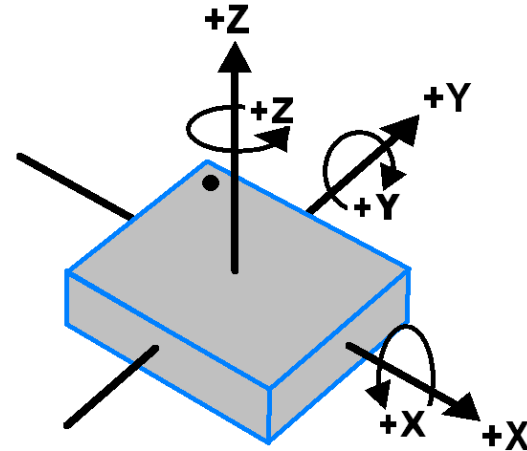
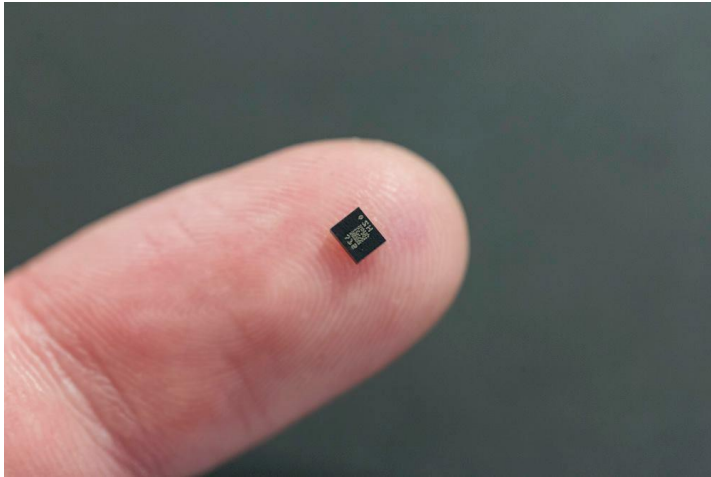
$$y(t) = \underbrace{\frac{2m\ddot{\theta}\omega X}{k}}_{\text{call } Y} \cos(\omega t)$$

$$\ddot{\theta} = \left(\frac{k}{2m\omega X} \right) Y$$



Inertial Measurement Unit - IMU

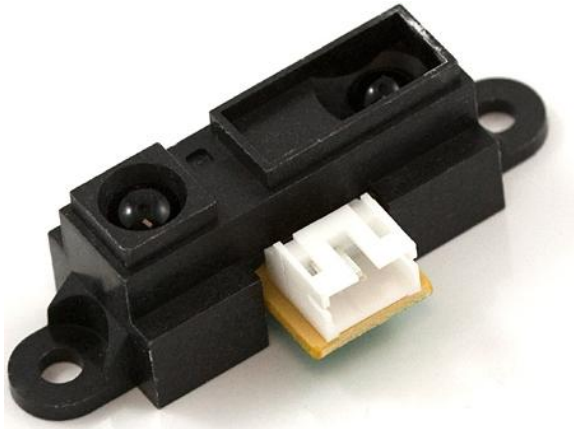
- An IMU is a compact unit that makes use of the inertial sensors
- Can measure linear accelerations and rotational rates along 3 axes
- Often include a magnetometer to provide a magnetic field reference
 - Helps with heading and orientation



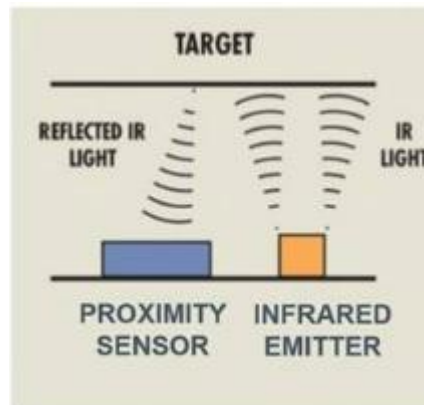
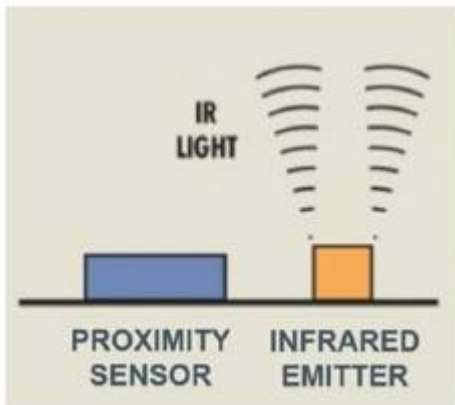
Take A Stretch Break



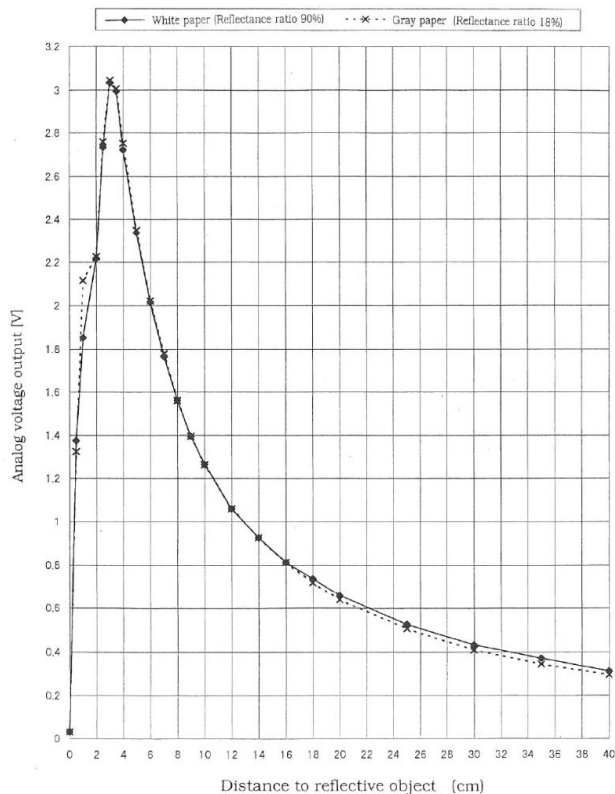
Infrared (IR) Proximity Sensor



- An IR LED (emitter) and an IR detector (photodiode) in the same package
- Sensor produces a voltage as a function of the intensity of IR light reflected back
- The intensity depends on the distance L the object is at (nonlinear relation)



Infrared (IR) Proximity Sensor



Electro-optical Characteristics

Parameter	Symbol	Conditions	MIN.	TYP.	MAX.	Unit
Measuring distance range	ΔL	(Note 1)	4	-	30	Cm
Output terminal voltage	V_o	L=30cm (Note 1)	0.25	0.4	0.55	V
Output voltage difference	ΔV_o	Output change at L change (30cm $\rightarrow$ 4cm) (Note 1)	1.95	2.25	2.55	V
Average supply current	I_{cc}	L=30cm (Note 1)	-	12	22	mA

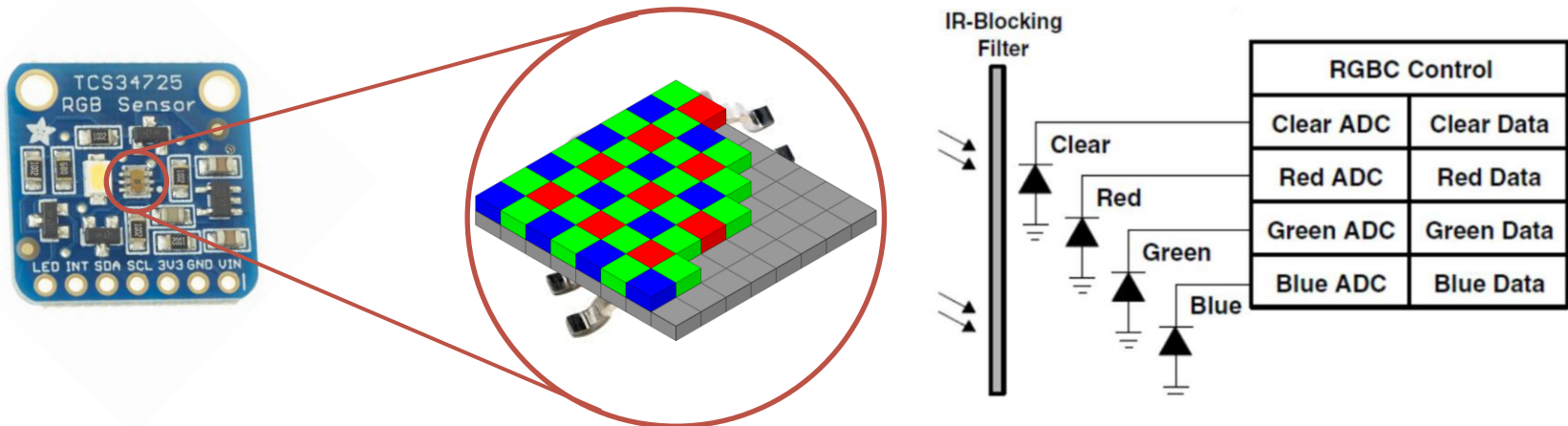
※L : Distance to reflective object

(Note 1) Using reflective object : White paper

(Made by Kodak Co., Ltd. gray cards R-27 • white face, reflective ratio : 90%)

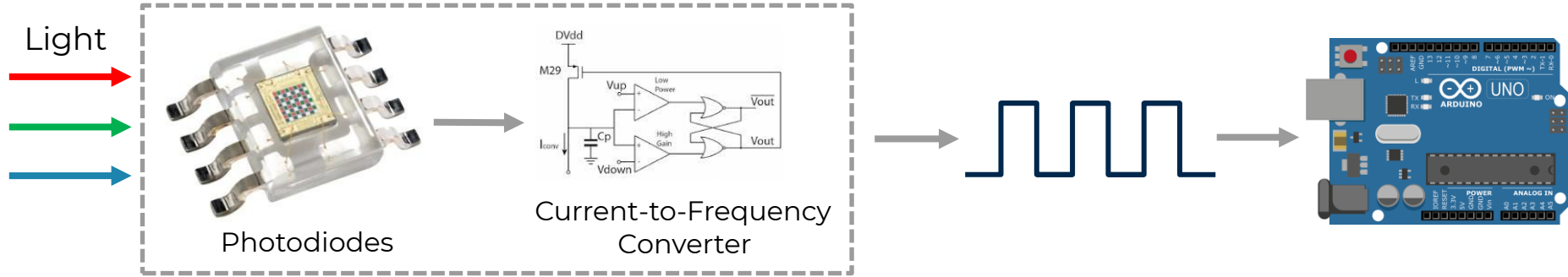
RGB Color Sensor

- RGB color sensors have an array consisting of red-, green-, and blue-filtered **photodiodes** as well as a clear (unfiltered) photodiode
- Some also include an **IR filter** to remove IR components of light to improve detection accuracy for visible light
- The intensity of each color results in an **analog** current for that color



RGB Color Sensor

TCS 3200 RGB Color Sensor

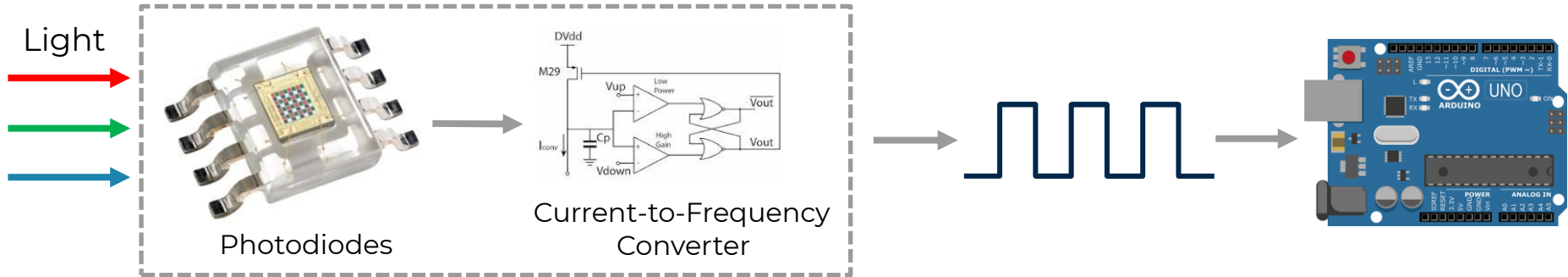


- The current for each color (RGB & clear) is converted into a two-value digital pulse train
 - This is a square wave with frequency directly proportional to the current
- Arduino inputs the square wave via its digital ports and uses a frequency counter to convert the square wave's frequency into an intensity value for each color

RGB Color Sensor

Given all of that, is the RGB color sensor an **analog** or **digital** sensor?

TCS 3200 RGB Color Sensor



Factors to Consider when Choosing a Sensor

- Range of Measurement
- Accuracy
- Repeatability/Precision
- Noise
- Sensitivity/Resolution
- Non-linearity
- Hysteresis
- Stability/Drift
- Bandwidth (Dynamic Characteristics)
- Cost/Complexity
 - Power Consumption
 - Post Processing



Range of Measurement



- The range of measurement defines the limits between which the input can vary and defines the largest value which can be measured
- Span is often used to report range of measurement and is the maximum value minus the minimum value
 - Example: The span of the IR proximity sensor shown below is 26 cm

■ Electro-optical Characteristics

Parameter	Symbol	Conditions	MIN.	TYP.	MAX.	Unit
Measuring distance range	ΔL	(Note 1)	4	-	30	Cm
Output terminal voltage	V_o	L=30cm (Note 1)	0.25	0.4	0.55	V
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※L : Distance to reflective object

(Note 1) Using reflective object : White paper

(Made by Kodak Co., Ltd. gray cards R-27 • white face, reflective ratio ; 90%)

Range of Measurement

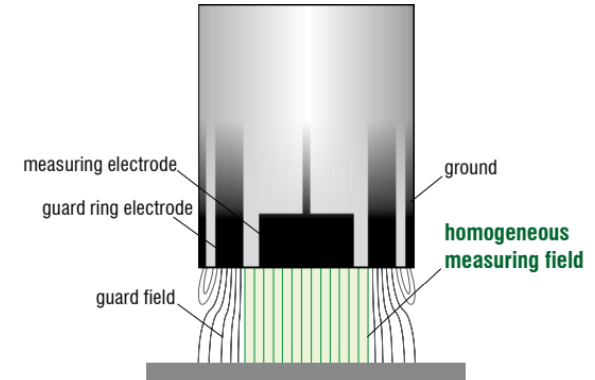
Linear Encoder
(large range > 1 m)



Linear Potentiometer
(small range ~100 mm)

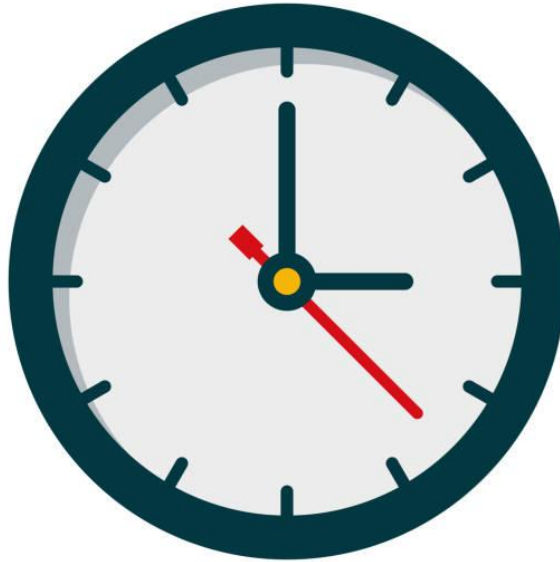


Capacitive Probes
(very small range ~100 μm)



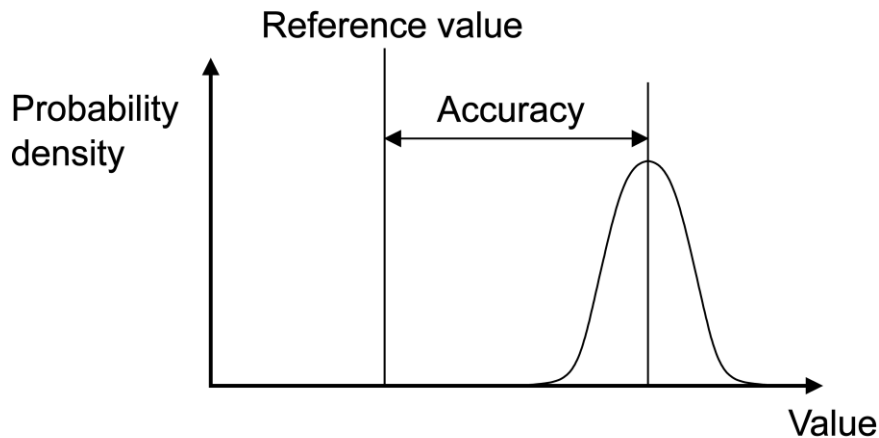
Discussion Question

Would you rather have a clock/watch that is always 60 minutes late or one that is sometimes 5 minutes late and at other times 5 minutes early?



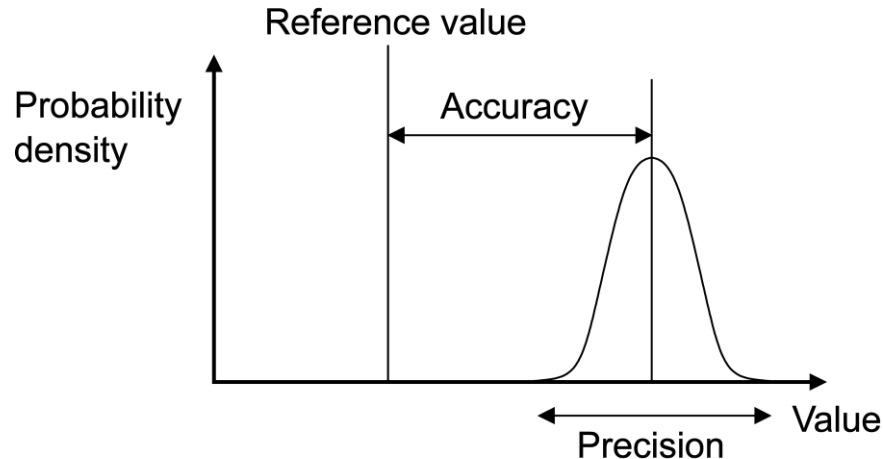
Accuracy

- Sensor **accuracy** is the measure of how close a sensor's reading is to the true or actual value it is trying to measure
 - A way of reporting “how wrong is my measure”



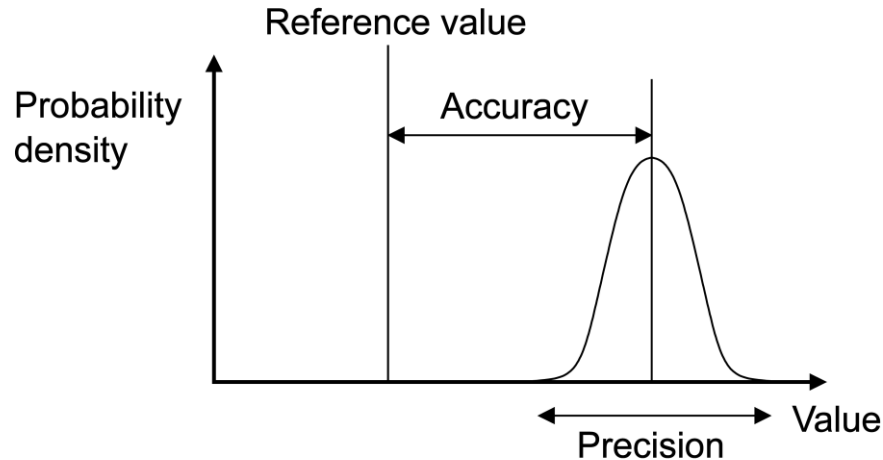
Repeatability/Precision

- The **repeatability** or **precision** of a sensor is the amount of variation in measurements
 - If they are taken by a single person using the same sensor on the same item and under the same conditions
- It is the degree to which repeated measurements under unchanged conditions show the same results



How do we Improve?

- If no sensor is perfect, how do we get around these short falls?
- Sensor **accuracy** can be improved by means of calibration
 - Clock being 60 minutes slow
- To reduce **precision** errors, average several measurements from the sensor
 - If the clock is sometimes 5 minutes fast or 5 minutes slow, the average would be right

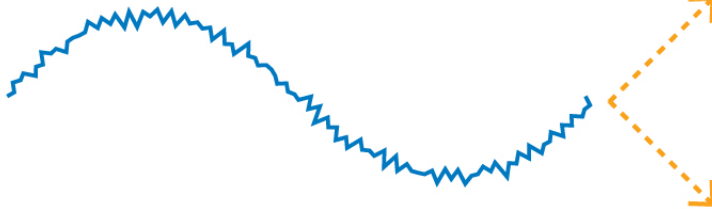


Sensor Noise

- Noise are **disturbances** to the sensor signal that reduce the repeatability
- The **signal-to-noise ratio** is the ratio of the power of the desired signal to the power of the undesired noise

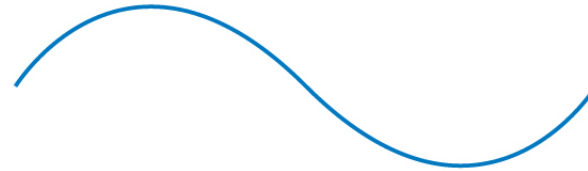
What we observe

SIGNAL + NOISE



Isolated noise from signal

SIGNAL



NOISE



Sensitivity & Resolution



For analog sensors:

- The **sensitivity** (S) is the ratio of the change in the measured quantity versus the change in signal being measured

$$S = \frac{\Delta Y}{\Delta X}$$

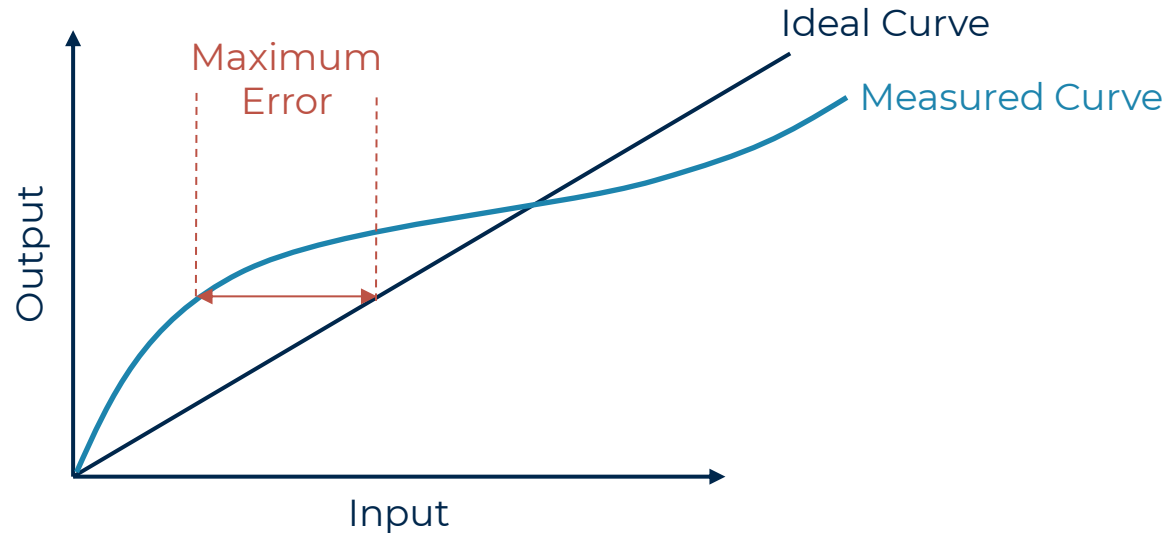
- A higher sensitivity means that the measured quantity can be resolved better (i.e. detect smaller changes)

For digital sensors:

- Digital sensors have the characteristic that the output will change in small step, so the **resolution** of a sensor is the smallest change in the input value that will produce an observable change in the output value
 - I.e. the base resolution of an optical encoder is 360 degrees/# of slots

Non-Linearity Error

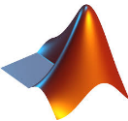
- For many sensors, we assume a linear relationship between input and output
- Unfortunately, few transducers have a truly linear relationship and thus errors occur because of this assumption
- **Non-linearity error** is defined as the maximum difference from the linear relationship



Stability/Drift

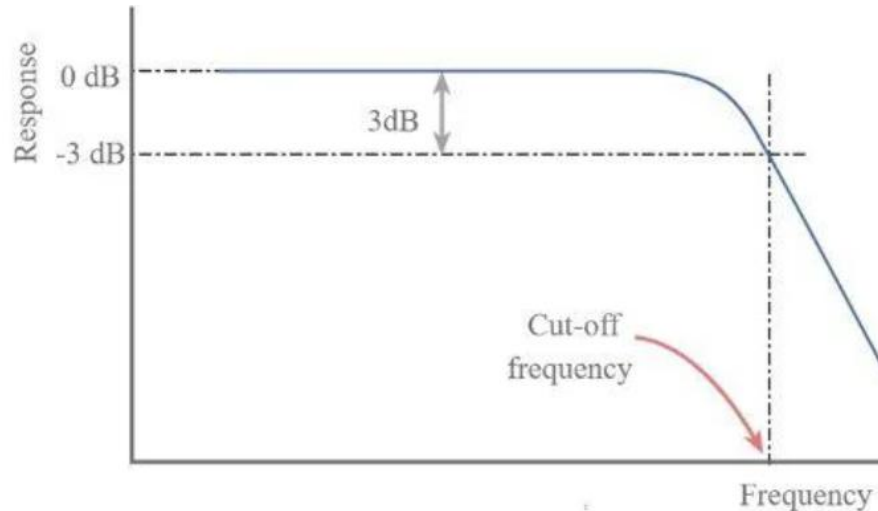


- The **stability** of a sensor is its ability to give the same output when used to measure a constant input over a period of time
- The term **drift** is used to describe a change in output that occurs over time



Bandwidth/Response Time

- The **bandwidth** is the frequency range over which the sensor provides a valid and useful output
- Similarly, the **response time** is the time the sensor needs to produce a valid output after the input has changed
- Higher bandwidth = faster response



Moving Forward...

- Let's look at signals!

